

Tower Bridge

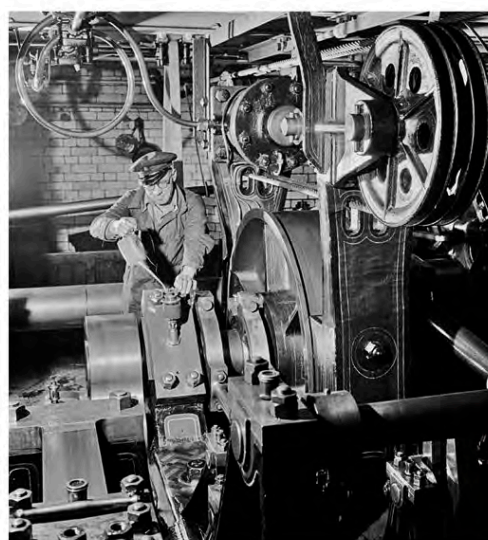
A Neo-Gothic bridge, masterfully engineered to allow the passage of shipping



Traffic on London's roads grew in the late 1800s, outstripping the capacity of the capital's bridges and highlighting the need for a crossing downstream of the congested London Bridge. However, a new bridge at such a location would need to allow the passage of merchant ships on their way up to the Pool of London—the wharves on either side of the river. In 1884, after much discussion, an ingenious design by engineer John Wolfe Barry (1836–1918) and architect Horace Jones (1819–1887) was approved, and its specifications were set out in an act of Parliament the following year. Its roadway is divided into two leaves known as bascules (from the French term for “balance scale”), which are lifted up to allow shipping to pass.

A grand opening

The bridge was built in the Neo-Gothic style fashionable at the time in order to harmonize with the nearby Tower of London. Construction began in 1886 and took 8 years. Two massive piers were sunk into the riverbed and filled with more than 78,400 tons (71,120 tonnes) of concrete to support the bridge's two towers and walkways, which were made of steel and clad in Cornish granite and Portland stone. The bridge was finally opened by Edward, Prince of Wales (later Edward VII), on June 30, 1894, and has been open to traffic—and raised for ships—ever since.



△ THE ENGINE ROOM

A workman oils the engines beneath Tower Bridge that pump steam into the hydraulic accumulators in order to raise or lower the bascules of the bridge.

▷ THE BRIDGE AT SUNSET

Tower Bridge is lit each evening to show off its many fine architectural features, particularly the brickwork of the two towers and the two upper walkways.



OPENING THE BRIDGE

When the bridge is opened, the weight of its leaves, or bascules, is counterbalanced by weights beneath each of the bridge's two towers. The power to raise the bascules originally came from two 360-horsepower steam engines that pressurized water in accumulators; the water was released to drive the lifting engines when needed. Opening and closing the bridge could take as little as 5 minutes. A new electrohydraulic drive system was installed in 1974.

